

FORUM:	United Nations Office for Outer Space Affairs
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STUDENT OFFICER:	Junwoo Lee
POSITION:	Deputy Chair of United Nations Office for Outer Space Affairs

Introduction

The rapid advancement in aerospace technology has enabled the launch of spaceships and artificial satellites by not only governments but also private companies. In 2024, the global aerospace industry is valued at \$361 billion, with projections reaching 734.5 billion by 2033. Alongside this growth, the issue of inequitable access to space technology and infrastructure by nations has emerged as a pressing international concern.



Figure 1. The Outer Space Treaty

Currently, only some advanced nations are able to access the outer space infrastructure and data and are capable of utilizing those for the benefit and in the interests of all nations, while other nations cannot. The inability to access satellite data directly isolates developing nations from effectively managing natural disasters, abating climate change, and securing provision and telecommunication. As such, this situation hinders achieving Article 1 of the Outer Space Treaty: the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries and shall be the province of all mankind. Thus, addressing this profound threat to global equality is of paramount importance. The inequality between access degrees to space technology and infrastructure has called for entirely new frameworks at the forefront of global policy discussions.

Background

Internationally, the Outer Space Treaty—since 1967—serves as the core legal instrument that prohibits the claim of sovereignty by means of use or occupation. However, the Outer Space Treaty in 1967 does not have concrete mechanisms or enforcement regarding how great powers should actually collaborate and share space technology with developing countries. Consequently, under the pretext of “space for all,” it resulted in only a few nations possessing the technology and occupying space.

With the immense commercial value of the Geostationary Orbits (GEO), which allow antennas to be fixed in alignment with the Earth’s rotation period, developed nations and private companies began to preemptively secure prime locations in Geostationary Orbits. Therefore, developing countries located near the equator, such as Colombia and Indonesia, asserted their sovereignty over the Geostationary Orbits above their borders and strongly opposed the orbital monopoly by developed nations through the 1976 Bogota Declaration. In response to the declaration, the International Telecommunication Union (ITU) applied the principle of “first come, first served.” Though this principle appears fair on the surface, in fact, it was an absolute advantage only for developed countries that possessed the technology and infrastructure to launch satellites.

As developments in outer space were made by developed countries, developing countries began to strongly demand rights to the tangible benefits provided by space technology such as remote sensing, communications, weather data, etc. Due to this pressure from developing countries, the Declaration on International Cooperation in the Exploration and Use of Outer Space for the Benefit and in the Interest of All States was adopted at the UN General Assembly—Space Benefits Declaration—in 1996. That said, space-faring nations had secured and even broadened a space technology gap. Ultimately, the accumulation of these historical inequalities wrought the present-day crisis that developing nations find themselves excluded from emerging fields and remain fully dependent on foreign infrastructure.

Problems Raised

Intensifying Space Technology Gap

The international bodies—despite the Outer Space Treaty of 1967 and Space Benefits Declaration of 1996—have failed to establish legally binding mechanisms that mandate or mitigate the space technology gap for developing nations. Consequently, these nations are systematically deprived of the opportunity to develop indigenous space infrastructure, such as satellite manufacturing capabilities and data processing technologies. This structural flaw leaves them to totally depend on the goodwill of space-faring nations or international bodies and the commercial interests of private corporations to access crucial satellite data.

This dependency severely hampers the ability of developing nations to address pressing domestic challenges, such as disaster management, climate change mitigation, and agricultural planning. Instead of

being active participants in space exploration and development, they are relegated to the role of mere consumers forced to purchase exorbitantly priced data from foreign entities. This commercialization of space data entrenches and exacerbates the global technological and economic segregation.

Lack of Local Infrastructure and Critical Brain Drain

Developing countries generally lack the crucial ground infrastructure required to utilize the data from space effectively, although this data is freely accessible or affordable. The absence of high-capacity data processing centers and advanced analytical software means that raw data cannot be translated into actionable intelligence. Without these localized facilities, the potential benefits of space technology—ranging from precision agriculture to early warning systems for natural disasters—remain largely untapped, rendering the data practically useless for local lawmakers.

This severe paucity in domestic infrastructure directly fuels a devastating brain drain phenomenon. The engineers, scientists, and researchers in developing countries observe the dearth of necessary facilities or funding to pursue advanced aerospace research in their own nations. Thus, the highly skilled personnel choose to migrate to established space agencies, such as NASA, and major private aerospace companies. This systemic loss of talent, human capital of developing nations, renders them in a perpetual cycle of technological reliance and underdevelopment.



Figure 2. The Sentinel-1 Flight Operations Control Centre in Darmstadt, Germany

International Actions

The Outer Space Treaty of 1967

Formalized in 1967, the Outer Space Treaty serves as the foundational legal cornerstone of international space law. Article I of the treaty explicitly mandates that the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic and scientific development, and shall be the province of all humankind. Furthermore, Article II reinforces this egalitarian principle by establishing that outer space—including the moon and other celestial bodies—is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means.

Nevertheless, it lacks concrete enforcement mechanisms, binding guidelines, or specific protocols mandating how space-faring nations should practically collaborate with developing nations. Therefore, under the banner of “space for all,” technologically advanced states were able to leverage their capital and industrial capabilities to preemptively secure key orbital positions and technological dominance, leaving developing nations without the practical means to exercise their rights guaranteed under the treaty.

The Space Benefits Declaration of 1996

In response to escalating demands from developing nations seeking equitable distribution of space technology and satellite data, the UN General Assembly adopted Resolution known as the Space Benefits Declaration. This sought to operationalize the broad principles of Article I of the Outer Space Treaty by explicitly requiring international space cooperation to consider the specific development needs of developing nations. It encouraged space-faring nations to foster capacity-building partnerships, facilitate data sharing, and assist developing countries in establishing indigenous space capabilities.

The Space Benefits Declaration, though, fell short of establishing a transformative framework for technology transfer due to its legal nature. It lacked mandatory legal force, leaving international cooperation entirely subject to the voluntary discretion and commercial interests of individual states. The declaration affirmed that states remain free to determine the terms of their international cooperation on a mutually acceptable basis, thereby allowing developed nations to protect their intellectual property rights and national security interests. Hence, the declaration failed to dismantle the commercial barriers facing developing countries, perpetuating their reliance on purchasing expensive foreign satellite services.

Key Players

International Telecommunication Union (ITU)

The International Telecommunication Union is the specialized United Nations agency responsible for allocating global radio- and satellite- spectrum and satellite orbits. In order to counter the monopolization of orbits by developed nations, the ITU has implemented a priori planning mechanism for certain Broadcasting-Satellite Service (BSS) bands. This system guarantees pre-allocated orbital slots and frequencies for all member states, including developing nations, ensuring that space is reserved for them even if they currently lack the technology and infrastructure to launch satellites. The ITU is currently paralyzed by an unprecedented regulatory crisis coordination request for over 1 million non-geostationary satellites. This overwhelming commercial rush by developed nations threatens to deplete available spectrum and exhaust the ITUs’ administrative capacity, making it exorbitantly difficult to guarantee remaining orbital slots for the future use of developing countries.

European Space Agency (ESA)

The European Space Agency is an influential intergovernmental space organization, comprising 22 European member states. It has historically distinguished itself by pioneering the “Open Data Policy,” acting as a crucial bridge for developing nations seeking access to advanced space technologies. ESA operates the Copernicus Programme, the world’s largest Earth observation initiative, which provides free, full and open access to data from its Sentinel satellites. To especially assist developing nations, ESA partnered with the African Union to launch the Global Monitoring for Environment and Security Africa (GMES & Africa) initiative, with 32 million euro in funding, to aid African institutions in utilizing satellite data for sustainable development and resource management. Those assists are boons for developing nations.

ESA faces the fundamental barrier of infrastructure inequality—despite the benevolent intentions of the open data policy—often described as “data is free, but access is not.”

The Copernicus Programme generates a staggering 250 terabytes of data daily. Developing nations lack the massive server capacities, high-speed broadband, and specialized algorithm engineers required to download and process this data, rendering the open-access policy practically inaccessible for many developing nations without further infrastructural aid.

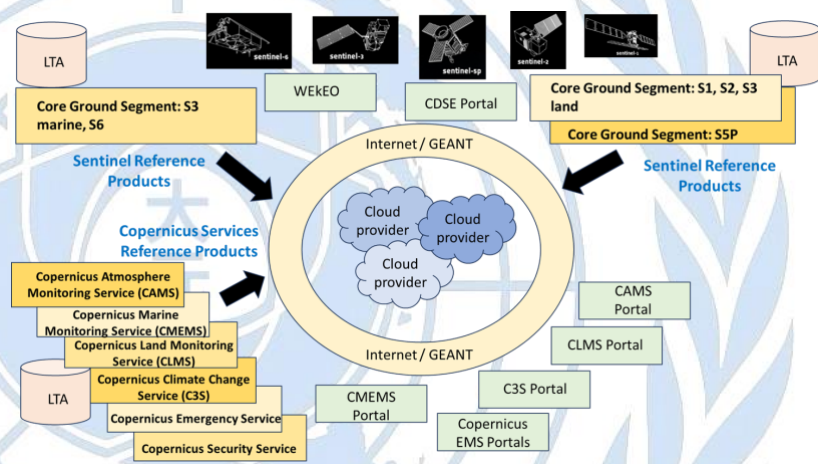


Figure 3. Diagram of the conceptual elements of Copernicus

United States of America

The United States of America plays a paramount role in shaping international space norms. Historically, the United States has championed free-market ideals and the “first-come, first served” principle to protect its technological advantage and foster commercial

innovation in space. The US government actively engages in capacity-building through programs. However, the USA faces severe diplomatic friction as it struggles to balance international demands for



Figure 4. Map of Global North and Global South

equitable access with the commercial interests of its domestic mega-corporations. The United States of America firmly opposes mandatory technology transfer treaties, citing national security risks and intellectual property (IP) protection. This perpetuates the technological gap between developed Global North and Global South.

Possible Solutions

Reform the ITU Orbital Allocation Framework

One option is to restructure the current “first-come, first-served” regulatory mechanism governed by the International Telecommunication Union (ITU). This approach could draw on establishing a standardized a priori planning system that explicitly reserves a specific proportion of Geostationary Orbit (GEO) slots and radio frequencies for developing countries. This proactive reservation system could be coupled with an international support mechanism, where entrenched space-faring nations provide technical assistance. Ultimately, this step-by-step cooperation would ensure that emerging nations have guaranteed, equitable access to space infrastructure once they gain the fundamental technological and financial capabilities in the future.

Establish an International Technology Transfer and Capacity-Building Fund

Another Approach is to create a UN-endorsed financial and technical mechanism dedicated to bridging the global space gap. This framework could draw inspiration from established space-faring nations, commercial aerospace mega-corporations, and international non-government organizations—such as the Secure World Foundation—to fundamentally subsidize the prohibitive costs of space infrastructure. By pairing free satellite data sharing with targeted capacity-building programs, this solution would empower developing nations to independently analyze space data for disaster management and agriculture. Ultimately, this localized infrastructure would help prevent severe brain drain and foster self-sustaining, indigenous space experts.

Key Terms

Geostationary Orbit (GEO)

A circular orbit 35,785 km (22,236 miles) above Earth’s Equator in which a satellite’s orbital period is equal to Earth’s rotation period of 23 hours and 56 minutes. That is, satellites in GEO remain fixed over a specific location. GEO is highly significant for telecommunications and weather monitoring.

Low Earth Orbit (LEO)

An orbit around Earth with a period of 128 minutes or less (making at least 11.25 orbits per day) and an eccentricity less than 0.25. LEO has become the center of geopolitical tension due to the rapid deployment of commercial mega-constellations.

Priori Planning

A priori planning means making a structured strategy, calculating sample sizes, or deploying operational resources before an event or data collection takes place.

Broadcasting-Satellite Service (BSS)

It is a radiocommunication service providing direct-to-home (DTH) television, audio, and broadband transmissions using high-power space stations for public reception.

Copernicus Programme

Copernicus is the Earth Observation component of the European Union's space programme, looking at our planet and its environment for the benefit of Europe's.



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